From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

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ALL INDIA TEST SERIES

JEE(Main)-2014

ANSWERS, HINTS & SOLUTIONS
CRT–II
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	**B**	**C**	**C**
2.	**A**	**B**	**A**
3.	**A**	**C**	**C**
4.	**A**	**C**	**B**
5.	**B**	**B**	**B**
6.	**C**	**B**	**C**
7.	**C**	**D**	**A**
8.	**C**	**B**	**D**
9.	**A**	**B**	**D**
10.	**C**	**A**	**A**
11.	**D**	**B**	**B**
12.	**A**	**C**	**A**
13.	**B**	**D**	**A**
14.	**C**	**A**	**B**
15.	**A**	**C**	**C**
16.	**D**	**C**	**D**
17.	**D**	**B**	**D**
18.	**A**	**C**	**C**
19.	**B**	**C**	**B**
20.	**A**	**B**	**C**
21.	**B**	**D**	**C**
22.	**B**	**C**	**D**
23.	**D**	**A**	**A**
24.	**B**	**C**	**D**
25.	**A**	**B**	**B**
26.	**A**	**D**	**C**
27.	**D**	**A**	**D**
28.	**B**	**A**	**D**
29.	**D**	**C**	**A**
30.	**C**	**B**	**A**

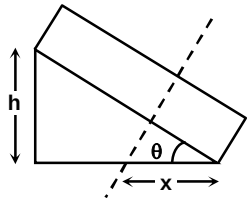
Physics

PART – I

SECTION – A

1. $\therefore$ Net force on the system is
$F = 3\,mg \sin\theta - \mu_k\, 3\,mg\cos\theta$
$\therefore \quad a_{cm} = \dfrac{3mg\sin\theta - 3\mu_k mg\cos\theta}{3m} = 4\ m/s^2$

2. Consider an arbitrary moment when the wedge has travelled a distance x into region II.
The area of the top surface inside the region II = ax sec θ
Force on it = ax sec $\theta\ \Delta P$ = ax sec θ [ΔP =1]
Component of the force opposite velocity = ax sec θ. sin θ = ax tan θ.
If it further moves by dx then the work done = ax tanθ dx



$\therefore \quad \dfrac{1}{2}mv_0^2 = a\tan\theta\int_0^b x\,dx$

$\Rightarrow \quad v_0 = \sqrt{\dfrac{abh}{M}}$

3. $S_A = v_0 t - \dfrac{1}{2}\mu_A g t^2$

$S_B = v_0 t - \dfrac{1}{2}\mu_B g t^2$

$S_A - S_B = L - \ell \Rightarrow t = \sqrt{\dfrac{2(L-\ell)}{(\mu_B - \mu_A)g}}$

4. $N/N_0 = e^{-\lambda t}$

5. First law of motion.

6. For vertical oscillation time period

$T_1 = 2\pi\sqrt{\dfrac{m}{k}}$ . . . (1)

For the transverse oscillation period of simple pendulum $T_2 = 2\pi\sqrt{\dfrac{\ell}{g}}$

But $2T_1 = T_2 \Rightarrow k = \dfrac{4mg}{\ell}$

8. $-\dfrac{3GMm}{2R} + \dfrac{1}{2}mv^2 = 0$
$\Rightarrow \quad v = \sqrt{3gR}$

9. Tension in the string $T = mg - F_B = 3\rho vg - \rho vg = 2\rho vg$
Balancing torque
$kxR/2 = TR$
$\Rightarrow \quad x = 2T/k = 4\rho vg/k$

10. II and IV quadrant field will be added.

11. $v_P = \dfrac{1600 \times 10^{-6}}{4 \times 10^{-4}} = 4$ m/s

$v_Q = \dfrac{1600 \times 10^{-6}}{2 \times 10^{-4}} = 8$ m/s

$\rho_A - \rho_B = \dfrac{1}{2}\rho_A\left(v_Q^2 - v_P^2\right) = \rho gh$

$h = 3.6$ mm

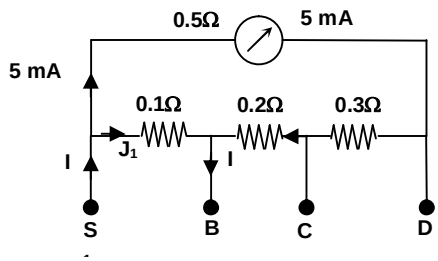
12. $mg - T = ma$ ----- (1)

$T_r = \dfrac{mr^2}{2} \times \dfrac{a}{r}$ ----- (2)

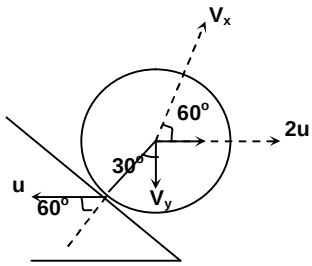
From (1) and (2), $a = \dfrac{2g}{3}$ & $T = \dfrac{mg}{3} = \mu mg$

$\mu = \dfrac{1}{3}$

13. $J_1 \times 0.1 = 5 \times I$
$J_1 = 50$ mA
$I = 50 + 5 = 55$ mA



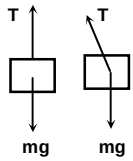
16. Let V_x and V_y be components of velocity. At contact point A and B,
the velocity along normal should be same.
$u\cos 60^\circ = V_y \cos 30^\circ - V_x \cos 60^\circ$ …. (1)
$v_x = 2u$
$v = \sqrt{V_x^2 + V_y^2}$



19. $qEy = \dfrac{1}{2}mv^2$
Magnetic force will not do any work.

20. Conceptual.

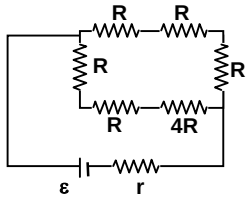
21. Now downward force on the right block is more.



22. $I_C = I_0 + M(OC)^2 = I_0 + M(OB^2 + BC^2) = I_B + M(BC)^2$

23. Coulombic force between them remains same.

24. Circuit is forming a wheatstone bridge $R_{eq} = 2R$
For maximum power transfer $2R = r$.



25. $\phi = \frac{\sqrt{3}}{4}\ell^2 B$

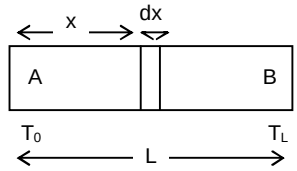
$\varepsilon = \left|\frac{d\phi}{dt}\right| = \frac{\sqrt{3}}{4}\ell^2\frac{dB}{dt}$

$i = \frac{\varepsilon}{R} = \frac{\sqrt{3}\ell^2}{4R}$

26. $C = \sqrt{\frac{\gamma RT}{M_0}} = \sqrt{\frac{5RT}{3M_0}}$

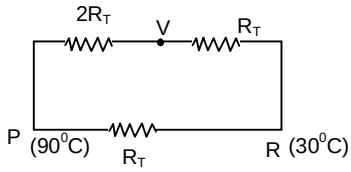
$dx = C.\,dt = \sqrt{\frac{5R}{3M_0}\left[T_0 + \left(\frac{T_L - T_0}{L}\right)x\right]}dt$

$t = \frac{2L}{(\sqrt{T_L} + \sqrt{T_0})}\sqrt{\frac{3M}{5R}}$



27. Equivalent circuit is

$\frac{90 - V}{2R_T} = \frac{20}{R_T} \Rightarrow V = 50^0C$



28. Leftward Force = $\frac{2I}{C} = \frac{\Delta P}{\Delta t}$

29. Conceptual, torque of pseudo force.

30. Effective length = $(r_2 + r_1)$

Chemistry

PART – II

SECTION – A

1. $P_1V_1^{\gamma} = P_2V_2^{\gamma}$

$\Rightarrow \frac{P_1}{P_2} = \left(\frac{V_2}{V_1}\right)^{\gamma} = \left(\frac{\rho_1}{\rho_2}\right)^{\gamma} = (32)^{\frac{7}{5}}$

$= \left(2^5\right)^{\frac{7}{5}} = 2^7 = 128$.

2.

	$NH_4Cl \rightleftharpoons$	$NH_3(g)$	+	HCl
$t = 0$	0.20	0		0
At eqm.	$(0.2 - x)$	x		x

$K_P = P_{NH_3} \times P_{HCl} = P^2 \Rightarrow P = \sqrt{K_P}$

$= \sqrt{0.36} = 0.6$ atm.

NH_3 formed $= \frac{PV}{RT} = \frac{0.6 \times 10}{0.0821 \times 600}$

= 0.122 mole = moles of NH_4Cl decomposed.

$\therefore$ Moles of NH_4Cl left = 0.2 – 0.122 = 0.078.

7. $Na_2B_4O_7 + \underset{(X)}{H_2SO_4} \longrightarrow Na_2SO_4 + H_2B_4O_7$

$H_2B_4O_7 + 5H_2O \longrightarrow 4H_3BO_3$

$2H_3BO_3 \xrightarrow{\Delta} B_2O_3 + 3H_2O$

$B_2O_3 + \underset{(X)}{6Na} \longrightarrow 2B + 3Na_2O$

8. $\Delta G^o = \Delta H^o - T\Delta S^o$

$H_2O \rightleftharpoons H^+ + OH^-$ $\quad \Delta H^o = 13.7$ kcal

$\Delta S^o = \frac{\Delta H^o - \Delta G^o}{T} = \frac{13.7 + 19.14}{298}$

= 0.1102 kcal mol^{-1} = 110.2 cal k^{-1} mol^{-1}

9. In bcc, $r = \frac{\sqrt{3}\,a}{4}$

Edge length = a

$\therefore$ Edge length not covered by atom = a – 2r

$= a - \frac{\sqrt{3}}{2} \times a = a\left(\frac{2 - \sqrt{3}}{3}\right)$

$\therefore$ percentage fraction not covered = $\frac{a\left[\frac{2-\sqrt{3}}{2}\right]}{a} \times 100 = 0.134 \times 100$ = 13.4%

10. $Al^{3+} + 3e \longrightarrow Al \qquad K = \dfrac{1}{\left[Al^{3+}\right]}$

$Al(OH)_3 \rightleftharpoons Al^{3+} + 3OH^-$

pH = 14, $[H^+] = 1 \times 10^{-14}$

$(OH^-) = 1$ M

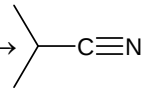
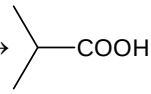
$K_{sp} = [Al^{3+}][1]^3$

$\therefore \left(Al^{3+}\right) = 1\times10^{-33}$ M

$E = E^o - \dfrac{0.0591}{3}\log\dfrac{1}{\left[Al^{3+}\right]}$

$= -1.66 - \dfrac{0.0591}{3}\log10^{33} = -1.66 - \dfrac{0.0591}{3}\times33$

= - 2.31 V

11. $CH_4 \xrightarrow[hv]{Cl_2} CH_3Cl \xrightarrow{H_3C\quad MgCl} CH_3CH_2CH_3 \xrightarrow{Br_2/hv}$  $\xrightarrow{KCN}$  $\xrightarrow{H_3O^+}$ 

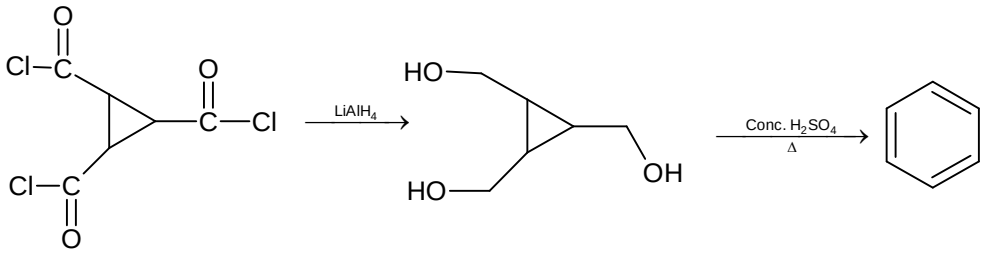
13. $\dfrac{M_1V_1}{n_1}\left(CO_3^{2-}\right) = \dfrac{M_2V_2}{n_2}\left(H^+\right)$

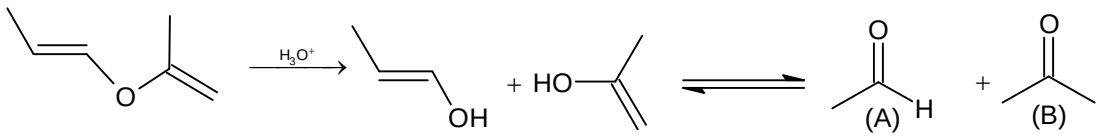
$\Rightarrow \dfrac{M_1\times50}{1} = \dfrac{4\times49.35}{2} \quad \therefore M_1 = 1.974$

$M = \dfrac{x\times d\times10}{m_B} \Rightarrow 1.974 = \dfrac{x\times1.25\times10}{106}$

$\therefore x = 16.7$

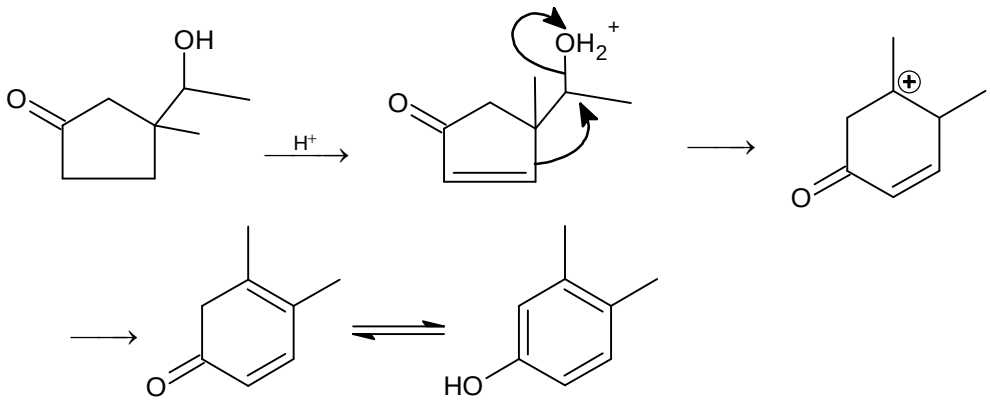
15. H_2O is weak field ligand while CN^- is a strong field ligand. In complex (i), the distribution is $t_{2g}^3e_g^2$, i.e. all electrons are unpaired while in (ii) complex, the distribution is $t_{2g}^5e_g^0$, i.e. two t_{2g} orbitals are paired while one is unpaired.

16. 

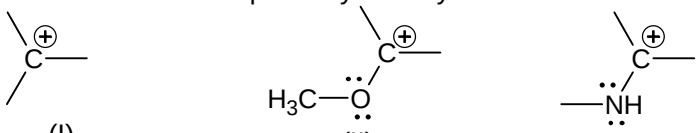
17. 

Aldehyde and ketone can be differentiated by Fehling's solution.

18.



19. The rate can be compared by stability of carbocation.



(I) (II) (III)

N has more tendency to donate electron than O.

21. $H = E + PV \Rightarrow H = E + nRT$

$H_2 = E_2 + nRT_2$

$H_1 = E_1 + nRT_1$

$(H_2 - H_1) = (E_2 - E_1) + nR(T_2 - T_1)$

$\Rightarrow \Delta H - \Delta E = nR\Delta T = 4 \times 8.314 \times 50$

= 1660 J

22. $H_2S \rightleftharpoons 2H^+ + S^{2-}$

$CdS \rightleftharpoons Cd^{2+} + S^{2-}$

$K_{sp_{CdS}} \rightleftharpoons \left[Cd^{2+}\right]\left[S^{2-}\right]$

$\rightarrow 8 \times 10^{-27} = 1 \times 10^{-14}\left[S^{2-}\right]$

$\therefore \left[S^{2-}\right] = 8 \times 10^{-13}$ M

$K_{H_2S} = \left[H^+\right]^2\left[S^{2-}\right]$

$\Rightarrow \left[H^+\right]^2 \times 8 \times 10^{-13} = 1 \times 10^{-22}$

$\therefore \left[H^+\right]^2 = 1.25 \times 10^{-10}$

$[H^+] = 1.11 \times 10^{-5}$

$\therefore$ pH ≈ 5 .

23.

$N_2O_4(g) \rightleftharpoons 2NO_2(g)$	
1	0
$(1-x)$	2x

Total moles = 1 – x + 2x = 1 + x

Since PV = nRT, $V \propto n$

% of $NO_2 = \dfrac{2x}{(1+x)} \times 100 = 50$

$\therefore x = \frac{1}{3}$

$(1+x) = \frac{D}{d} = \frac{\text{Theoretical VD}}{\text{Experimental VD}}$

$\Rightarrow \frac{4}{3} = \frac{46}{d} \Rightarrow d = \frac{46 \times 3}{4} = 34.5$

28.

Species	NO^-	NO^+	CN^-	CN^+
Bond order	2	3	3	2

Bond order $\propto \frac{1}{\text{Bondlength}}$

Bondlength of NO^- > CN^+ due to greater number of antibonding electrons.

29. $\Delta T_f = i \times K_f \times m \Rightarrow 86 \times 0.001$

$\Rightarrow 0.0054 = i \times 1.86 \times 0.001$

$\therefore i = \frac{0.0054}{1.86 \times 0.001} = 2.9 \approx 3$

i.e. there are 3 ions in the solution of complex.

So, $[Pt(NH_3)_4Cl_2]Cl_2$ is the correct choice.

Mathematics

PART – III

SECTION – A

1. $\left|\frac{2(3a+4b+c)+3(4a+3b+c)}{\sqrt{a^2+b^2}}\right| = 10$

$\Rightarrow \left|\frac{18a+17b+5c}{\sqrt{a^2+b^2}}\right| = 10$

$\Rightarrow \left|\frac{\frac{18}{5}a+\frac{17}{5}b+c}{\sqrt{a^2+b^2}}\right| = 2$

2. $b^2 = 4 \times 2 = 8,\ 2ae = \sqrt{40}$

$a^2e^2 = 10 = b^2 + a^2$

$\Rightarrow a^2 = 2$

Centre = (1, 5)

Equation of transverse axis $\equiv x - 3y + 14 = 0$

Equation of conjugate axis $\equiv 3x + y - 8 = 0$

Equation of hyperbola = $\frac{(3x+y-8)^2}{1} - \frac{(x-3y+14)^2}{4} = 20$

3. $f(x) = \frac{x^4 - 14x^2 + 25}{x - \frac{5}{x} + 2} = \frac{x^2\left(x^2 + \frac{25}{x^2} - 14\right)}{x - \frac{5}{x} + 2}$

$$= \frac{x^2\left(\left(x-\frac{5}{x}\right)^2-4\right)}{x-\frac{5}{x}+2} = x^2\left(x-\frac{5}{x}-2\right)$$

$\Rightarrow x^2 - 2x - 5 = 0$

$$x = \frac{2\pm\sqrt{4+20}}{2} = \frac{2\pm 2\sqrt{6}}{2} = 1\pm\sqrt{6}$$

4. 6 places can be selected in ${}^9C_6$ ways and 6 can be placed only at 5 places, except the right most of other 6 selected. Remaining numbers i.e. 7, 8, 9 in 3! ways

Hence number of ways = ${}^9C_6 \cdot 5 \cdot 3!$

5. $z = 18 + 26i$, $r = 10\sqrt{10}$, $\tan\theta = \frac{13}{9}$, $\theta \in \left(0, \frac{\pi}{2}\right] \Rightarrow \frac{\theta}{3} \in \left(0, \frac{\pi}{6}\right]$

$$\therefore\ z^{1/3} = \left(10\sqrt{10}(\cos\theta + i\sin\theta)\right)^{1/3} = \sqrt{10}\left(\cos\frac{\theta}{3} + i\sin\frac{\theta}{3}\right)$$

$$\tan\theta = \frac{3t-t^3}{1-3t^2},\ \left(t = \tan\frac{\theta}{3}\right)$$

$$\Rightarrow \tan\frac{\theta}{3} = \frac{1}{3}$$

$z^{1/3} = 3 + i$, $a = 3$, $b = 1$

6. Let parabola is $y^2 = 4ax$

$\alpha = at_1t_2$, $\beta = a(t_1 + t_2)$, $Q = (at_1t_3, a(t_1 + t_3))$, $R \equiv (at_2t_3, a(t_2 + t_3))$

Let T = (h, k) then $h = \frac{at_3}{2}(t_1 + t_2) = \frac{\beta t_3}{2}$

$$2k = a(t_1 + t_2) + 2at_3 = \beta + \frac{4ah}{\beta}$$

The locus id $4ax - 2\beta y + \beta^2 = 0$

7. We observe that $\theta = \phi$

because of perpendicularity of two secant

Now $\angle SOR + \angle SOQ - \angle SOP = \pi$ $(\theta = \phi)$

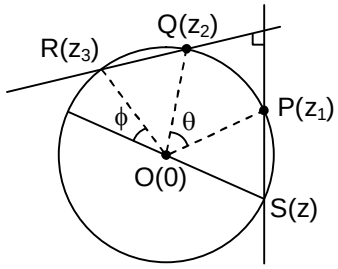
$\Rightarrow (\arg(z_3) - \arg(z) + (\arg(z_2) - \arg(z)) - (\arg(z_1) - \arg(z)) = \pi$

$\Rightarrow \arg(z_3) + \arg(z_2) = \pi + \arg(z_1) + \arg(z)$

$= \arg(z_2z_3) = \pi + \arg(zz_1)$

$\Rightarrow z_2z_3 = -zz_1$

$\Rightarrow zz_1 + z_2z_3 = 0$



8. $$C_n = \frac{1}{n}\int_{\frac{n}{n+1}}^{1} \frac{\tan^{-1}(t)}{\sin^{-1}(t)}dt$$

$$L = \lim_{n\to\infty} n^2 \cdot C_n = \lim_{n\to\infty} n\int_{\frac{n}{n+1}}^{1} \frac{\tan^{-1}t}{\sin^{-1}t}dt\ \ (\infty \times 0)$$

$$L = \lim_{n\to\infty} \frac{\int_{n}^{\overline{(n+1)}} \frac{\tan^{-1}t}{\sin^{-1}t}dt}{\frac{1}{n}}$$ (Applying Leibniz rule)

9. We have to find non negative integral solution of equation $2x + y + z = 21$
Note that x can take a maximum value of 10 and a minimum value of 0.
We rearrange equation so that we get an integer equation with y and z as variables, x as a constant
$y + z = 21 - 2x$
The number of non negative integer solution is ${}^{21-2x+2-1}C_1 = {}^{22-2x}C_1 = 22 - 2x$
We now add the number of solutions so obtained for all the possible values of x.
The total number of solution is therefore $\sum_{x=0}^{10}(22-2x) = 132$

10. As $a > 0 \Rightarrow$ R.H.S. > 0 hence $x > 0$
Applying AM $\geq$ GM we get,
$$\frac{x a^{\frac{1}{x}} + \frac{1}{x}a^{x}}{2} \geq a^{\left(x+\frac{1}{x}\right)\frac{1}{2}}$$
L.H.S. $\geq 2\ a^{\frac{1}{2}\left(x+\frac{1}{x}\right)}$
But $x + \frac{1}{x} \geq 2\ \forall\ x > 0$
$\Rightarrow 2\ a^{\frac{1}{2}\left(x+\frac{1}{x}\right)} \geq 2\ a\ \forall\ a \geq 1 \Rightarrow$ L.H.S. $\geq 2\ a$ and equality holds if $x = 1$.

11. The required probability is $p = \dfrac{\frac{1}{4}\times\frac{{}^{12}C_2}{{}^{51}C_2}}{\frac{1}{4}\times\frac{{}^{12}C_2}{{}^{51}C_2} + \frac{{}^{13}C_2}{{}^{51}C_2}} = \frac{11}{50}$

12. $(x+2)^2 = 3\left(y+\frac{13}{3}\right)$ = latus rectum = 3
Other conic is $\frac{(x-3)^2}{7^2} + \frac{(y+2)^2}{\left(\frac{7}{2}\right)^2} = 1$ which an ellipse is
Latus rectum = $\frac{2b^2}{a} = \frac{2\cdot 49}{4\cdot 7} = \frac{7}{2}$
Positive difference $\frac{7}{2} - 3 = \frac{1}{2}$

13. $B = \text{adj}(2A) \Rightarrow |B| = |2A|^{n-1}$
$1024 = 4^{n-1}\ 2^{n(n-1)}$
$2^{10} = 2^{n^2+n-2}$
$n = 3$

14. 2f(x) and $\frac{f(x)}{2}$ have the same domain as f(x), f(2x) and f(x + 2), $f\left(\frac{x}{2}\right)$ have the range as f(x)

m = 2, n = 3

Verify by $f(x) = \sin^{-1} x$, d : $|x| \le 1$, R : $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right)$

15. $|a_1 \times a_2| = |\vec{a}_1||\vec{a}_2|\sin\frac{2\pi}{n}$

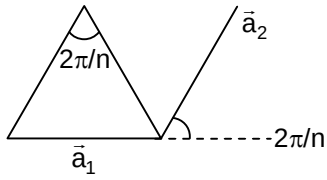
$|\vec{a}_1 \cdot \vec{a}_2| = |\vec{a}_1||\vec{a}_2|\cos\frac{2\pi}{n}$

$n\sin\frac{2\pi}{n} = n\cos\frac{2\pi}{n}$

$\tan\left(\frac{2\pi}{n}\right) = 1 \Rightarrow \frac{2\pi}{n} = m\pi + \frac{\pi}{4}$

$n = \frac{8}{1+4m}$

Smallest possible value of n corresponds to m = 0 $\Rightarrow$ n = 8



17. $f^{-1}(x) = [x] + \sqrt{\{x\}}$

$A = \int_{1/2}^{5} [x] + \sqrt{\{x\}}dx$

$= \int_{1/2}^{5} [x]dx + \int_{1/2}^{1} \sqrt{\{x\}}dx + 4\int_{0}^{2} \sqrt{\{x\}}dx = \frac{1}{3\sqrt{2}}\left(40\sqrt{2} - 1\right)$ sq. unit

18. $\tan^2 \alpha \to a$, $\tan^2 B \to b$

$$\frac{(a+1)^2}{b} + \frac{(b+1)^2}{a} = \left[\frac{a^2}{b} + \frac{1}{b} + \frac{b^2}{a} + \frac{1}{a}\right] + 2\left(\frac{a}{b} + \frac{b}{a}\right) \ge 4\left[\frac{a^2}{b} \cdot \frac{1}{b} \cdot \frac{b^2}{a} \cdot \frac{1}{a}\right]^{1/4} + 2 \cdot 2\left[\frac{a}{b} \cdot \frac{b}{a}\right]^{1/2} = 8$$

19. $$A = \frac{1}{2}\begin{vmatrix} 0 & 0 & 1 \\ 0 & \frac{c}{b} & 1 \\ \frac{c}{a} & 0 & 1 \end{vmatrix} = \frac{1}{2}\left[0 - \frac{c^2}{ab}\right]$$

If A is independent of a, b, c then $C^2 = ab$ a, c, b are G.P.

20. Using cosine rule $b^2 - 2bc\cos A + (c^2 - a^2) = 0$

$b_1 + b_2 = 2c\cos A$ $b_1 b_2 = c^2 - a^2$

Using $b_2 = 2b_1$

$b_1 = \frac{2c}{3}\cos A$ and $b_1^2 = \frac{c^2 - a^2}{2}$

$\frac{4c^2}{9}\cos^2 A = \frac{c^2 - a^2}{2} \Rightarrow 9(c^2 - a^2) = 8c^2\cos^2 A$

$\Rightarrow \frac{c}{a}\sqrt{1 + 8\sin^2 A} = 3$

22. $x^3 - x^2 - x - 2 = 0 \Rightarrow (x - 2)(x^2 + x + 1) = 0 \Rightarrow x = 2$ … (1)

$x y^2 + 2 x y + 4 x - 2 y^2 - 4 y - 8 = 0 \Rightarrow (x - 2)(y^2 + 2 y + 4) = 0 \Rightarrow x = 2$ … (2)
Both the equation represent same line. So number of triangle formed are zero.

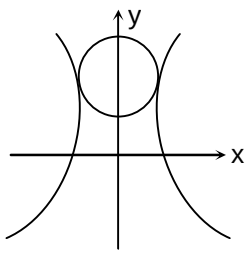
23. $f(x) = \sin^3 x + \lambda \sin^2 x$
$f'(x) = 3 \sin^2 x \cos x + 2 \lambda \sin x \cos x$
for exactly one minimum and one maximum
$f'(x) = 0$ at two points in $\frac{-\pi}{2} < x < \frac{\pi}{2}$
therefore $f'(x) = 0 \Rightarrow \cos x = 0$ or, $\sin x (3 \sin x + 2 \lambda) = 0 \Rightarrow \sin x = \frac{-2\lambda}{3}$ or $\sin x = 0$, $\cos x \neq 0$
$\Rightarrow -1 < \frac{-2\lambda}{3} < 1$
$\Rightarrow \lambda < \frac{3}{2}$ but $\lambda > 0 \Rightarrow 0 < \lambda < \frac{3}{2}$

24. $P(x) = ax^2 + bx + c$
$P(x) = Q_1(x) + 4 = Q_2(x + 1) + 3 = Q_3(x - 1) + 1$
$P(0) = c = 4$
$P(-1) = a - b + 4 = 3$ $(b - a = 1)$
$P(1) = a + b + 4 = 1$ $(b + a = 3)$
$b = -1$, $a = -2$
$P(x) = 2x^2 - x + 4$
$P(2) = -8 - 2 + 4 = -6$

25. Center $(-3, -5)$
Radius $\rightarrow 2$
For L_1 to be a chord its distance from center must be less than radius
Since P is an integer p can take value $S_1 = \{17, 18, 31\}$
L_2 to be chord of a circle, P can take value from the set $S_2 = \{11, 12, \ldots.. 25\}$
$S_1 \cup S_2$ has 21 elements while
$S_1 \cup S_2$ has 9 elements
Required probability $\frac{9}{21}$ or $\frac{3}{7}$

26. $\lim_{n\to\infty} \frac{1}{\frac{(x-2)^n}{3^n} + 3 - \frac{1}{n}}$ divided N and D by $3^n \cdot n$
$\lim_{n\to\infty}$ to equal to $\frac{1}{3} \Rightarrow \lim_{n\to\infty} \frac{1}{n} \rightarrow 0$
$\lim_{n\to\infty} \left(\frac{x-2}{3}\right)^n \rightarrow 0 \Rightarrow -1 < x < 5$

27. Let the equations of circle be
$x^2 + (y - \lambda)^2 = 16$
$\lambda = \pm 2\sqrt{6}$



28. $|x + y + z| = |x| + |y| + |z|$
$\Rightarrow x^2 - 2 < 0, \sin x < 0$
$x \in (-\sqrt{2}, 0)$

29. The normal to the plane will have direction cosine $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and the distance of this plane from the origin will be OP = $\sqrt{3}\lambda$

$\frac{x}{\sqrt{3}} + \frac{y}{\sqrt{3}} + \frac{z}{\sqrt{3}} = \sqrt{3}\lambda = x + y + z = 3\lambda$

Intercepts lengths $(3\lambda, 3\lambda, 3\lambda)$

$(3\lambda)^2 + (3\lambda)^2 + (3\lambda)^2 = k^2$

$\lambda = \pm\frac{k}{\sqrt{3}}$

30. $\frac{x^2 + e}{x^2 + 1} \in (1, e]$, $f(x) = \ln\left(\frac{x^2 + e}{x^2 + 1}\right) \in (0, 1]$

$g(\theta) = \sqrt{\sin\theta} + \sqrt{\cos\theta}$ where $\theta \in (0, 1]$

$g'(\theta) = \frac{\cos\theta}{2\sqrt{\sin\theta}} - \frac{\sin\theta}{2\sqrt{\cos\theta}} = 0 \Rightarrow \theta = \frac{\pi}{4}$

$g(\theta)$ has an extremum point $\theta = \frac{\pi}{4}$

$g(0) = 1 \Rightarrow g(1) = \sqrt{\sin 1} + \sqrt{\cos 1}$

While $g\left(\frac{\pi}{4}\right) = 2^{3/4} \Rightarrow g(1) > g(0) = 1$

Since $g\left(\frac{\pi}{2}\right) = g(0) = 1$

$g(x) = \left(1, 2^{3/4}\right]$